

JONATHAN KIM

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Education

The University of Texas at El Paso (UTEP)

Expected: December 2026

Bachelor of Science in Computer Science

GPA: 3.73

Relevant Coursework: Machine Learning, Data Structures & Algorithms, Computer Organization, Software Engineering

Extracurriculars: Undergraduate Teaching Assistant; Secretary, Free Open Source Software Club

Experience

Crescent Electric Supply Company

December 2025 – Present

Software Engineer (Contract)

El Paso, Texas

- Streamlined blueprint takeoffs using Detectron2 + OCR, cutting estimation from 2 weeks to 8 minutes.

PM Technologies LLC

November 2025 – Present

Systems Engineering Intern

El Paso, Texas

- Engineered autonomous drone navigation using SLAM + Intel RealSense to automate factory 3D printer monitoring.
- Constructed YOLOv8-based 3D print defect detection to halt faulty builds and reduce material waste.

ReconAIssance

October 2025 – Present

Co-Founder & ML Engineer

El Paso, Texas

- Implemented aerial facial recognition using SCRFD + ArcFace on Jetson Orin Nano for real-time edge identification.
- Devised YOLOv8 weapon detection using Open Images V7 to identify firearms and knives in real time.
- Integrated autonomous drone tracking using MAVSDK, PX4, and Intel RealSense for vision-driven target following.

The Ohio State University AI-EDGE Institute

May 2024 – July 2024

Generative AI Researcher

Columbus, Ohio

- Optimized Stable Diffusion 2 training using PyTorch data pipelines to increase training throughput.
- Reduced generative model size using pruning and quantization to lower inference latency.
- Enhanced model generalization using dataset augmentation and preprocessing.

Projects

AI VTuber | Python, PyTorch, LLaMA-3, Whisper, Piper TTS, RAG, Concurrency

May 2024 – Present

- Designed real-time human-AI VTuber control using LLaMA-3, Whisper, and Piper TTS for shared avatar control.
- Synthesized context-aware AI conversations using RAG and persistent memory for multi-turn dialogue.

Sigan | Flutter, Dart, Gemini API, TensorFlow Lite, SQLite, NLP

May 2025 – November 2025

- Created a voice-first scheduling assistant using Gemini API and NLP to extract tasks from speech.
- Formulated a behavioral learning scheduling engine to optimize task placement.

Fight Coach (Hackathon) | Python, MediaPipe, OpenCV, Hugging Face Transformers, Flask

April 2025

- Detected boxing and exercise form errors from pose sequences.

DUI Risk Radar (Hackathon 2nd Place) | JavaScript, React, Firebase, HTML/CSS

September 2024

- Predicted high-risk DUI zones using spatiotemporal patterns.

Vocowbularly.courses (MLH HackWesTX) | React, Node.js, MongoDB, Gemini API

September 2024

- Generated phoneme-level pronunciation feedback from speech audio.

Poké Generator | Python, PyTorch, Stable Diffusion, LoRA

May 2024 – July 2024

- Refined text-to-image generation quality via model fine-tuning.

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, SQL, C

Frameworks/Libraries: PyTorch, OpenCV, MediaPipe, Pandas, Flask, React, React Native, Node.js

Tools/Platforms: Git, Firebase, AWS, Docker, REST APIs